

VEER ARORA

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PROFILE

Backend and AI engineer. I build systems that report their own reliability — evaluation harnesses with held-out splits, policy gates on agent actions, and tenant isolation enforced in the database rather than in application code. Validated an 8-model production MLOps platform at L&T Finance on GCP and Kubeflow. Recent work includes a RAG system where recall@5 went from 0.37 to 0.77 against a measured 0.85 ceiling, and a multi-tenant platform whose access control is enforced by PostgreSQL row-level security. B.Tech Computer Science & Design, 2026.

SKILLS

Languages & Data Python, SQL, Pandas, NumPy, data cleaning & validation, EDA

Backend & Web FastAPI, PostgreSQL, row-level security, SQLAlchemy, Alembic, REST APIs, React, TypeScript, Docker, pytest, Git, CI/CD

AI & LLM LLM tool-calling agents, RAG, LangGraph, evaluation harnesses (tool-selection, task-success, grounding, cost/latency), OpenAI / Anthropic / Google Gemini APIs

Analytics & ML correlation analysis, descriptive statistics, outlier & anomaly detection, scikit-learn

Visualization & Reporting Power BI, Matplotlib, Seaborn, PDF report generation

Cloud & MLOps Google Cloud Platform, Kubeflow, pipeline testing, UAT

Tools Streamlit, SQLite, n8n, OpenCV, MS Excel, MS Office

EXPERIENCE

L&T Finance — Data Analyst Intern

Jun 2025 – Aug 2025

- Validated Nostradamus, an 8-model MLOps platform on GCP and Kubeflow spanning EWS, Banking, Self-Cure and Collections workflows, ahead of production release.
- Caught an output-formatting defect in the EWS model that left its output column blank downstream — a failure that shows up as missing data rather than as an error — and drove root-cause analysis with the developers and MLOps engineers who owned the pipeline.
- Wrote SQL verification across pipeline stages: data-integrity and row-level checks that separated “the pipeline ran” from “the pipeline produced the right numbers”.
- Ran end-to-end pipeline validation and UAT in Python and SQL — test planning, defect tracking, deployment-readiness review.

E-Cell, IIT Bombay — Campus Executive

Aug 2023 – Feb 2024

- Promoted entrepreneurship programs, workshops, and competitions, and coordinated outreach, event logistics, and mentor coordination across the student community.

PROJECTS

AgentOps — AI Support Engineer

github.com/veer0608/agentops

Python · FastAPI · LangGraph · SQLAlchemy · Alembic

- Built an AI support agent that takes real actions (create/update tickets, escalate) behind a policy gate that auto-executes, routes to a human review queue, or blocks each action based on business rules and confidence thresholds; writes are idempotent and recorded in an append-only audit log.
- Made the model provider-agnostic — OpenAI, Anthropic, and Google Gemini swappable via one config value — and implemented a second agent on LangGraph behind the same interface, using the eval harness to prove both implementations score identically.
- Wrote a scenario-based evaluation harness scoring six dimensions (tool-selection precision/recall/F1, task-success, escalation accuracy, citation-grounding, cost, and latency), persisting every run for before/after tracking.
- FastAPI + SQLAlchemy + Alembic (SQLite dev / Postgres); 33 automated tests; read tools include RAG-backed doc search and GitHub-issue lookup.

CiteRAG — Retrieval-Augmented QA with Golden-Set Eval

github.com/veer0608/citerag

Python · FastAPI · SQLite-vec · SQLAlchemy

- Built a retrieval-augmented QA system over messy 10-K PDFs where every answer is grounded in cited source chunks.
- Created a golden-set eval harness (recall@k, MRR) and lifted recall@5 from 0.37 to 0.77 against a measured ceiling of 0.85 by iterating on chunking strategy and adding a cross-encoder re-ranker — measured, not guessed.
- Local embeddings + reranker with SQLite-vec; runs with no paid API required.

schemablind — Schema-Blind SQL Agent

github.com/veer0608/schemablind

Python · SQLite · BIRD benchmark · pytest

- Built a SQL agent given no schema: it explores a database it has never seen through four read-only verbs — list tables, describe, sample rows, run query — and composes its own answer, including learning from its own failed queries.
- Scored on BIRD execution accuracy: the agent's rows against the reference query's rows on the same database, with no judge model deciding whether SQL “looks right”.
- Held 40% of questions back as a test split assigned by hash of the question text, so a prompt tuned on failures cannot choose what it is marked on.
- Proved the harness before trusting it — an oracle solver must score 100% and a mute solver 0%, asserted in CI on every push. Sandboxed model-composed SQL behind four independent layers, with CI asserting the database is byte-identical after attempted writes. 119 tests.

AgencyDesk — Multi-Tenant SaaS Platform

github.com/veer0608/agencydesk

React · FastAPI · PostgreSQL · Docker

- Built a multi-tenant client/project management platform where one deployment serves many isolated agencies, each with its own clients, projects, and client portal.
- Enforced tenant isolation and internal-vs-client visibility with PostgreSQL row-level security and composite foreign keys, so cross-tenant access is impossible even if application code is wrong; the API connects as an unprivileged database role.
- Wrote 70 automated tests proving cross-tenant isolation and role-based visibility; containerized with Docker Compose and added GitHub Actions CI that builds and tests on every push.

EDUCATION

Madhav Institute of Technology & Science, Gwalior — B.Tech, Computer Science & Design

2022 – 2026

Shanti Niketan HS School, Vidisha — Class 12 (2021) · Trinity Convent Sr. Sec. School, Vidisha — Class 10 (2019)

CERTIFICATIONS

- Affective Computing — NPTEL (Jan – Apr 2024)
- Digital Design with Verilog — NPTEL (Jan – Apr 2025)
- Rational Thinking — Udemy

LEADERSHIP & ACTIVITIES

- R&D Coordinator, Startup Cell, MITS (Jan 2023 – Jun 2026)
- Core Team Member, Digital Learning Group, MITS (Feb 2023 – Jul 2024)
- State-level swimmer

LANGUAGES

English, Hindi